

Python Data Stack — Pop Quiz

Python · Pandas · Seaborn · Matplotlib

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Python

1. What does the following code print?

```
x = [1, 2, 3]
y = x
y.append(4)
print(x)
```

- ☐ A) [1, 2, 3]
- ☒ B) [1, 2, 3, 4]
- ☐ C) [4]
- ☐ D) Error

Python

2. Which creates a dictionary mapping each number to its square?

```
nums = [1, 2, 3, 4]
```

- ☒ A) {n: n**2 for n in nums}
- ☐ B) [n: n**2 for n in nums]
- ☐ C) {n**2: n for n in nums}
- ☐ D) (n: n**2 for n in nums)

Python

3. What is the output? return b

```
def f(a, b=[]):
    b.append(a)
    print(f(1))
    print(f(2))
```

- ☒ A) [1] then [2]
- ☐ B) [1] then [1, 2]
- ☐ C) Error
- ☐ D) [1, 2] then [1, 2]

Pandas

4. What does `df.groupby('category')['sales'].sum()` return?

- A) A single integer total
- B) A Series with categories as index and summed sales as values
- C) A DataFrame with a 'sum' column
- D) A list of sales values

Pandas

5. Which method fills missing values with the column mean?

```
df['col'].____(df['col'].mean())
```

- A) `replace()`
- B) `fillna()`
- C) `dropna()`
- D) `impute()`

Pandas

6. What is the key difference between `.loc` and `.iloc`?

- A) `.loc` filters by dtype, `.iloc` by column name
- B) `.loc` uses label-based indexing, `.iloc` uses integer position-based indexing
- C) `.loc` is faster, `.iloc` is slower
- D) They are identical — just aliases

Pandas

7. What does this produce?

```
pd.merge(df1, df2, on='id', how='left')
```

- A) Only rows present in both DataFrames
- B) All rows from `df2`, matching rows from `df1`
- C) All rows from `df1`, matched with `df2` where available
- D) All rows from both DataFrames

Seaborn

8. Which function best visualizes the distribution of a single continuous variable?

- A) `sns.scatterplot()`
- B) `sns.histplot()` or `sns.kdeplot()`
- C) `sns.barplot()`
- D) `sns.lineplot()`

Seaborn

9. What does the hue parameter do in most Seaborn plots?

- A) Changes the overall color palette
- B) Groups data by a categorical variable and maps each group to a different color
- C) Sets the transparency (alpha) of plot elements
- D) Selects which column to use as the y-axis

Seaborn

10. What does sns.FacetGrid do?

- A) Creates a single plot with multiple color layers
- B) Creates a grid of subplots, each showing a subset of the data
- C) Applies a grid background to an existing plot
- D) Stacks multiple DataFrames into one

Matplotlib

11. What is the critical gotcha with plt.show() vs plt.savefig()?

- A) plt.show() saves the file; plt.savefig() displays it
- B) plt.show() renders to screen; plt.savefig() writes to disk
- C) They are identical
- D) plt.savefig() must be called before plt.show() or the saved file will be blank

Matplotlib

12. What does this code produce?

```
fig, axes = plt.subplots(2, 3)
```

- A) A single plot with 6 data series
- B) 6 separate figures
- C) A figure with a 2-row, 3-column grid of subplots
- D) An error — subplots() takes only one argument

Matplotlib

13. How do you add a legend to a Matplotlib plot?

- A) plt.legend_add()
- B) plt.label(show=True)
- C) plt.legend() after setting label= on each plot call
- D) Legends are added automatically